

$$\begin{array}{c}
 f: M \rightarrow M \rightarrow \text{surjection} \\
 \uparrow \quad \quad \quad (\text{injection}) \\
 \text{Noetherian (Artin)} \rightarrow f: \text{iso}
 \end{array}$$

Artinian ring

$$d_{cc}$$

$$\mathbb{Z} \supset (2) \supset (2^2) \supset (2^3) \supset \dots$$

$$M/\ker f \cong M$$

$$\ker T^n \cap \text{Im } T^n$$

$$\ker T^n$$

$$0 \xrightarrow{f} L \xrightarrow{f} M \xrightarrow{g} N \rightarrow 0$$

$$0 \rightarrow L \xrightarrow{f} M \hookrightarrow N \rightarrow 0$$

$$M \xrightarrow{g} N \rightarrow 0 \Leftrightarrow g: \text{sur}$$

$$M: \text{f.f.} \Leftrightarrow L, N: \text{f.f.}$$

(Artin) (Artin)

$$M/f(L) \cong N$$

$$\bigoplus L \cong N$$

$$\text{(Cor)} M_i: \text{f.f.} \quad (1 \leq i \leq n)$$

$$\Rightarrow \bigoplus_{i=1}^n M_i: \text{f.f.}$$

$$0 \rightarrow M_1 \rightarrow \bigoplus_{i=1}^n M_i \rightarrow \bigoplus_{i=1}^{n-1} M_i \rightarrow 0$$

$T: V \rightarrow V$ is diagonalizable $\Leftrightarrow m_T$ splits and separable

" triangulizable $\Leftrightarrow m_T$ is separable

E/F is separable $\Leftrightarrow \# \text{Emb}_F(E, \bar{F}) = [E:F]$

E/F is splitting $\Leftrightarrow \text{Emb}_F(E, \bar{F}) = \text{Aut}(E/F)$

relation $\{ \}$

Thm. Let V be a finite-dimensional vector space, let $T: V \rightarrow V$ be linear. There is $k \in \mathbb{N}$ such that $V = \text{Im } T^k \oplus \text{Ker } T^k$.

Proof.